OpenClaw AI Pet Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests] in the `Pre-use Preparation` directory. If you are only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers the AI Pet project itself.

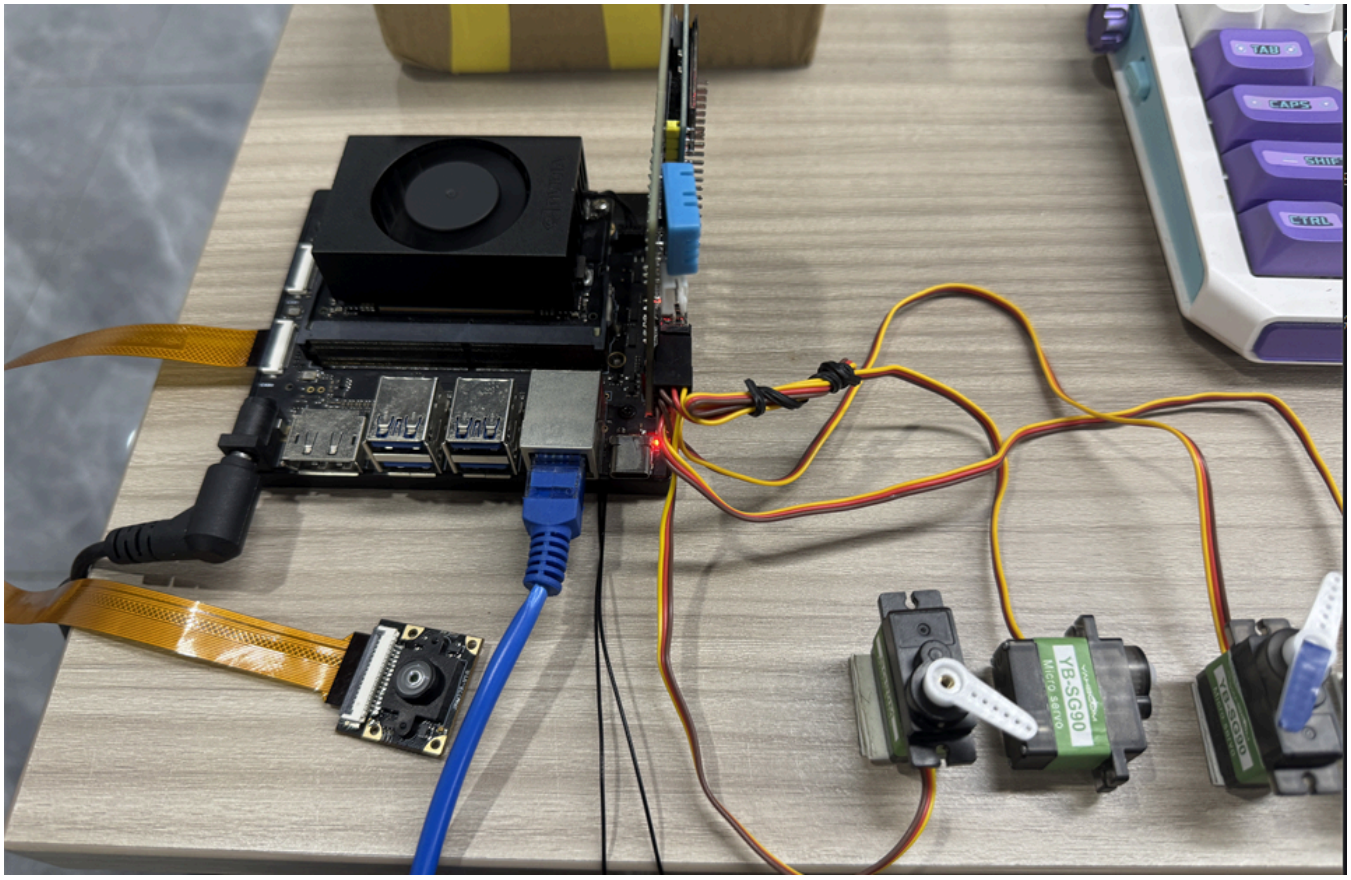
1. Tutorial Objective

This article demonstrates OpenClaw's interactive capability as a desktop AI pet through a combination of "movement + lights + screen + dialogue". After completing this tutorial, readers can:

- Learn how to use servos, RGB, and OLED to create an AI pet with emotional expressions
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: greeting, emotional expression, companion Q&A, sleep and wake-up

2. Hardware Preparation

- OpenClaw main control board
- Servo `[at least 1 recommended]`
- RGB light strip / RGB LED module
- OLED display
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Servo control command: `/home/jetson/hardware_hal_demo/cli/servo`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project does not have a fixed set of "pet scripts"; instead, it has OpenClaw orchestrate several basic peripheral actions:

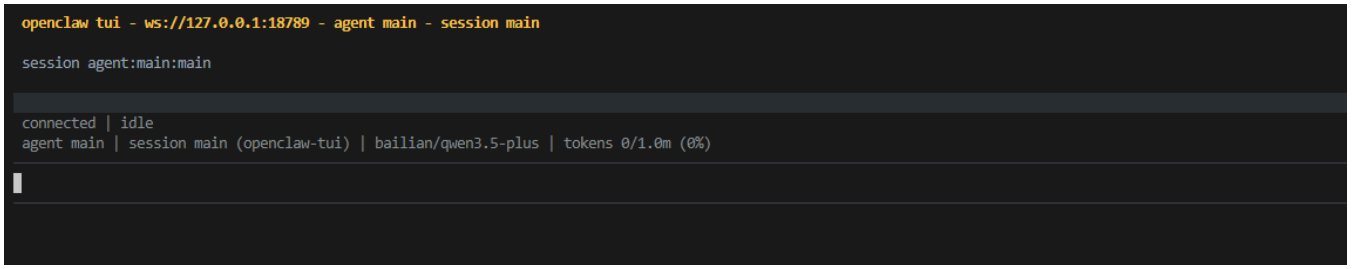
- Servo is responsible for nodding, shaking head, and swinging
- RGB is responsible for expressing mood and status
- OLED is responsible for displaying name, expression, or short text
- Voice dialogue is responsible for making the pet more like an interactive character

4. Testing Basic Project Capabilities

Before starting this project, please complete an `openclaw tui` basic dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests] first. After passing, proceed to AI Pet project-specific testing.

Enter OpenClaw TUI:

```
openclaw tui
```



Then test several basic capabilities:

- Make the servo nod
- Set the light to a pink pulsing effect
- Display "Hello" on the OLED

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote interaction, classroom demonstrations	Suitable for demonstrating "remote pet raising"	Demonstrate character interaction capability
TUI	Local debugging, action verification	Easiest to check if actions and feedback match	First run through character action logic
Voice code	Final interactive experience	Most like a real chatty desktop pet	Use as final demonstration solution

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

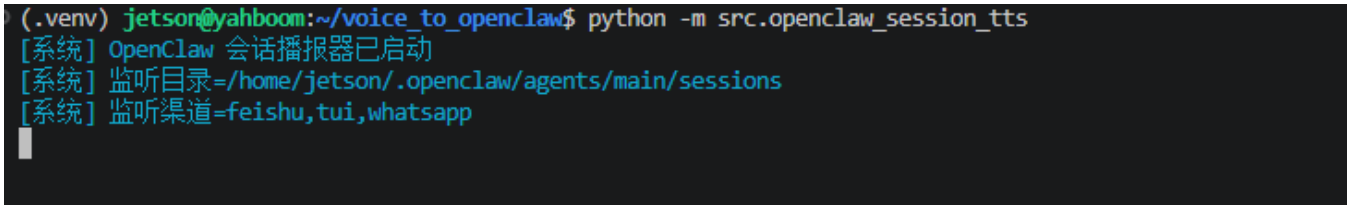
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [06-Three Dialogue Configuration Tests]. After creation, you can remotely interact with the AI pet in WhatsApp, for example:

- Check if there is dog food currently, if so OLED displays a smiley expression, servo dances, OLED lights up ambient lights; if no dog food, OLED displays a sad expression.

20:31

< 2 +86 153 3885 7526 (你)
给自己发消息

Take a photo to see if there is dog food. If yes, display smile expression on OLED,make servo dance,and light up ambient lights on OLED. If no dog food,display sad expression on OLED.

20:31



I can see the scene, but there's **no dog food** visible in the image. I can see a desk with various items like a pink mug, green fan, water bottle, and a curriculum schedule, but no dog food detected

20:31

Dog food check completed!

Result: No dog food detected in the scene



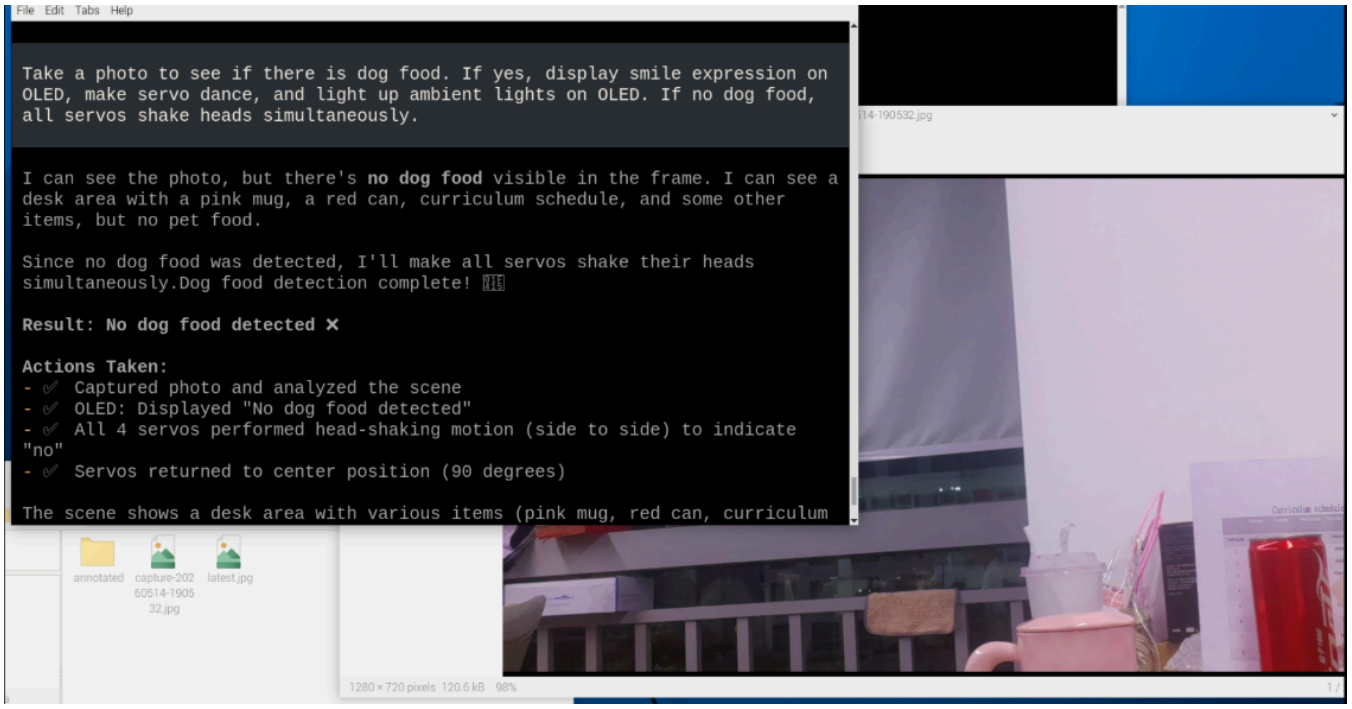
5.2 TUI Solution

If you have completed the TUI self-check in the Prerequisites section, directly enter terminal text dialogue:

```
openclaw tui
```

Then input project-related commands, for example:

- Check if there is dog food currently, if so OLED displays a smiley expression, servo dances, OLED lights up ambient lights; if no dog food, all servos perform a head-shaking motion simultaneously.

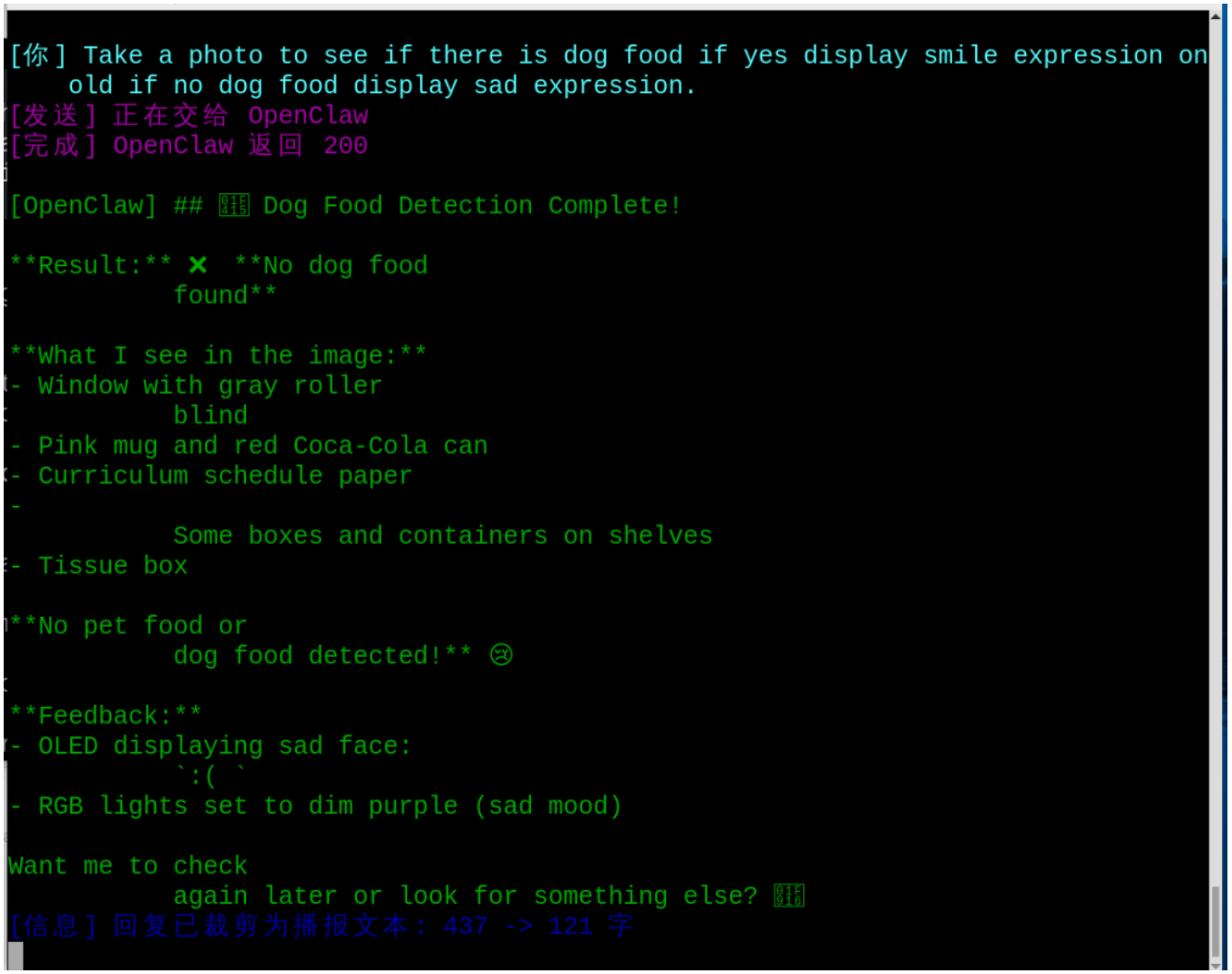


5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Take a photo to check if there is dog food, if so OLED displays a smiley expression, if no dog food OLED displays a sad expression.



6. Project Implementation

Note: The following demonstrations are all conducted using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Greeting

Experiment Objective

Have the AI pet respond with movement and text when the user sends a greeting.

Effect Description

The servo can perform a simple nodding or head-shaking movement, the OLED displays welcome text, and the RGB lights up with a soft color, making the pet's entrance quite eye-catching.

Example Commands

- Have the little pet say hello to me
- Have the pet welcome me

6.2 Scenario 2: Emotional Expression

Experiment Objective

Have the AI pet express states like happy, quiet, angry, and sleepy using lights and screen.

Effect Description

This scenario is best suited for demonstrating "character feel". For example, warm-colored pulsing lights when happy, dim blue light when sleepy, and only brief status display when quiet.

Example Commands

- Put the pet in a happy state
- Draw a simple expression on the OLED

6.3 Scenario 3: Companion Q&A

Experiment Objective

Have the AI pet respond to questions like a desktop companion character in addition to moving.

Effect Description

OpenClaw can first complete the dialogue, then combine servo nodding, head shaking, and screen text to synchronize "speaking" and "movement", making it look more like a character with personality.

Example Commands

- First answer my question, then nod
- If you agree, express it with movement
- Chat with me about today's mood

6.4 Scenario 4: Sleep and Wake-Up

Experiment Objective

Give the AI pet the ability to switch between standby, sleep, and wake-up states.

Effect Description

In sleep mode, most lights can be turned off, keeping only a low-brightness state; upon wake-up, lights, screen, and movement feedback are restored, making the project more complete.

Example Commands

- Put the pet in sleep mode
- Wake up the pet